

Prediction of Diabetes Patients Using MLP Neural Network

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1. Introduction

Diabetes prediction is an important task that facilitates the identification of individuals at risk for developing serious complications. Early prediction plays a crucial role in supporting timely medical intervention and improving patient outcomes (World Health Organization, 2023). In this project, a real-life dataset is analyzed to classify whether a patient does or does not have diabetes (Outcome = 1 or Outcome = 0). The dataset has eight clinical features: Pregnancies, Glucose, Blood Pressure, Skin Thickness, Insulin, BMI, Diabetes Pedigree Function, and Age that are commonly associated with diabetes risk. The dataset is divided into two sub datasets: The train dataset includes 615 records, and the test dataset includes 155 records. The goal is to train a MLP model to learn patterns from the training data and predict diabetes accurately from predictions on the test dataset.

2. Method

Prior to the development of the predictive model, exploratory data analysis was performed through the application of histograms and boxplots in order to get the distribution of features and identify abnormal values. This showed that Glucose, Blood Pressure, Skin Thickness, and BMI had values of zero which is not possible in actual clinical values. There were 4 and 11 zeros of glucose and BMI respectively; Blood Pressure and Skin Thickness had 211 zeros. Deleting 211 rows would drastically decrease the size of the dataset and would increase the chances of overfitting, so the zero values in Blood Pressure and Skin Thickness were replaced with the average of non-zero values instead. The correlation heatmap was also investigated, and it proved that the most significant predictors were Glucose and BMI of the outcome of diabetes. Thus, the two features were only subjected to outlier removal through a class-wise Interquartile Range (IQR) approach to maintain the size of datasets but emphasize on the most significant predictors. Lastly, z-score was used to index all features and Stratified sampling was used to divide the resulting cleaned dataset into 80 - 20 % .

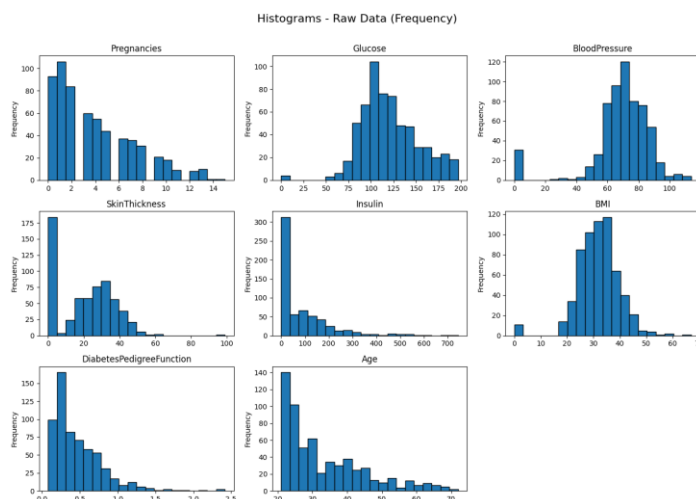


Fig 1: Histogram of the raw data

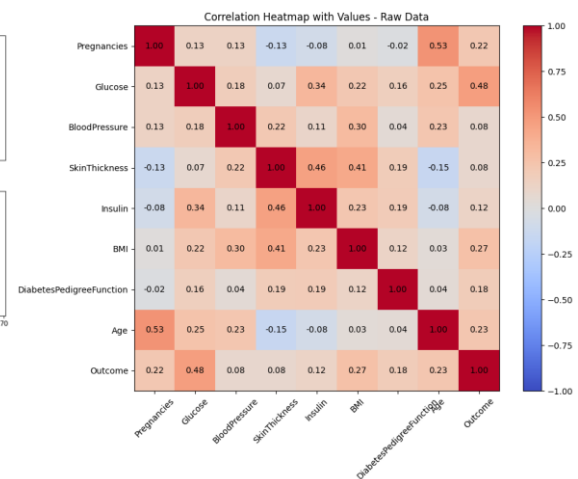


Fig 2: Correlation Heatmap

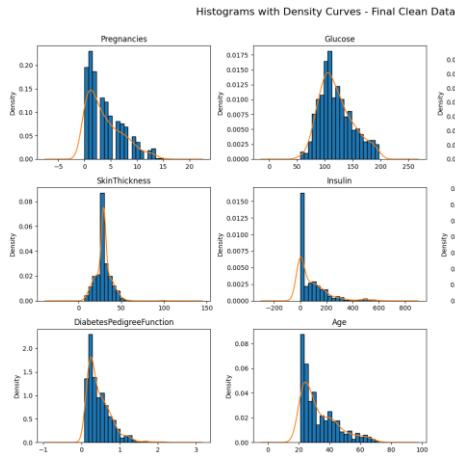


Fig 3: Density curve for the cleaned data

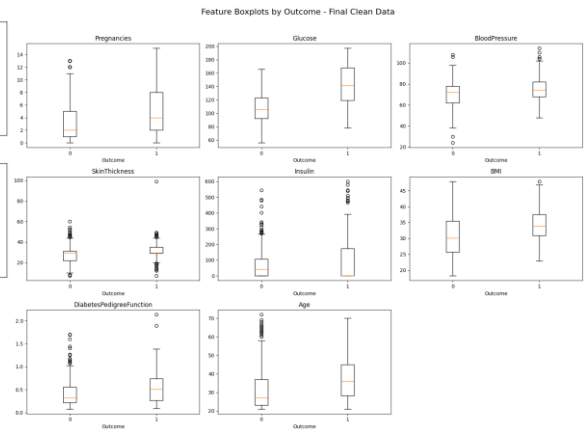


Fig 4: Boxplot for the cleaned dataset

The Multi-Layer Perceptron (MLP) neural network was created in PyTorch to classify diabetes into a binary. The model structure was made of an input layer containing eight neurons (one neuron per feature), two fully connected hidden layers containing 64 and 32 neurons apiece. The reason behind using Rectified Linear Unit (ReLU) activation was that it is faster to converge and avoids the problem of vanishing gradient. A dropout ($= 0.2$) was used after every hidden layer to minimize overfitting by randomly killing neurons throughout training. The output layer had only one neuron that gave the raw score (logit), which was transformed in a probability by a sigmoid function.

The model was trained with the Adam optimizer and the learning rate of 0.001. This optimizer assists the model in learning and varying weights. The loss package was Binary Cross-Entropy with Logits Loss (BCEWithLogitsLoss). It is stable and already has a sigmoid functioning. The rate of misclassifying instances of diabetes was made more expensive to the model by the addition of a class-weight. This assists in enhancing identification of diabetic patients. Training was made smoother with mini-batch learning that has a batch of 32. The early stopping of 10 epochs of patience was used to control overfitting. Activation based on ReLU assisted the model in learning more quickly. The data could not be memorised by the network because of the dropout. This dataset needed two hidden layers.

Three MLP configurations were experimented on the predictive performance to evaluate the impact of hyperparameters on the predictive performance which included hidden layer size, dropout, and learning rate.

3. Experimental Analysis

In this section, a detailed analysis of three Multi-Layer Perceptron (MLP) setups that are aimed at the prediction of diabetes is provided. Post-processing resulted in the 579 samples and 463 samples and 116 samples were used as training and validation samples, respectively, by splitting the dataset stratified to maintain the balance of the classes. The reason is that the classes were moderately weighted hence class weighting was introduced in the loss. It was trained in a batch size of 32, Adam, and early stopping to avoid overfitting.

There were several evaluation measures used: accuracy, precision, recall, F1-score, and the area under the Receiver Operating Characteristic curve (AUC). General correctness is represented by accuracy. Precision will gauge the number of times the predicted diabetics become diabetic. Recall is used to determine the level of capturing the diabetic people. F1-score is a balance between the

precision and the recall and the AUC is a measure of the probability ranking ability across the thresholds. In the case of medical applications, AUC is crucial, as the inability to identify a case of diabetes may be life-threatening.

3.1 Validation Performance Comparison

Three MLP settings were tested by varying hidden layer sizes, learning rate, and dropout rate. Table 1 summarises the validation set results:

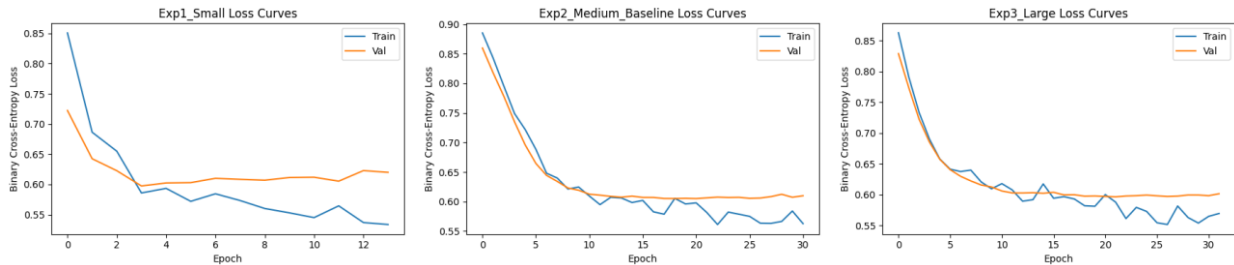


Fig 5: Loss curve of the experiments

Table 1. Validation Results for MLP Models

Model	Hidden Layers	LR	Dropout	Accuracy	Precision	Recall	F1-Score	AUC
Exp1_Small	32 → 16	0.01	0.2	0.7931	0.6429	0.9000	0.7500	0.8655
Exp2_Baseline	64 → 32	0.001	0.2	0.7845	0.6415	0.8500	0.7312	0.8632
Exp3_Large	128 → 64	0.0005	0.3	0.7759	0.6346	0.8250	0.7174	0.8704

Experiment 3 had the best AUC (0.8704) of the three models where it's accuracy, precision, recall and F1 are also seemed good. AUC is the most significant measure in medical classification problems since it gauges the model to differentiate diabetic and non-diabetic patients at all the probability thresholds. This would imply that Experiment 3 is most powerful in terms of discrimination and offers the most reliable predictions in terms of probabilities - a prerequisite of clinical decision support. Even though Experiment 1 achieved a slightly higher accuracy and recall, it has a low AUC, which implies lower risk ranking and worse threshold flexibility.

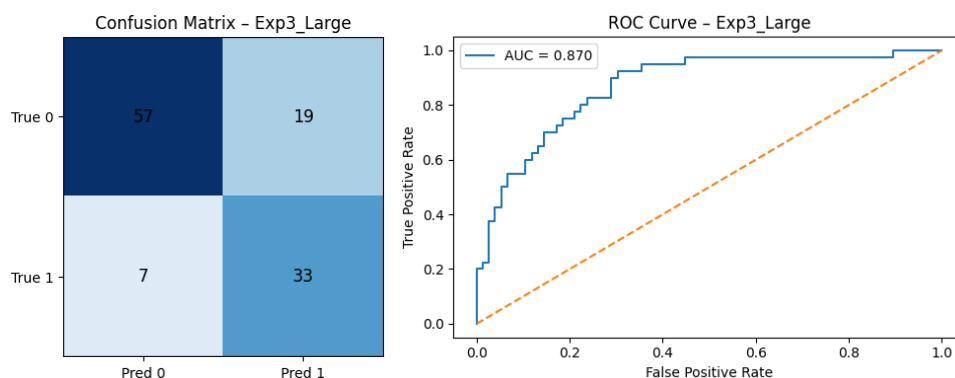


Fig 6: Confusion Matrix and ROC curve of Experiment 3

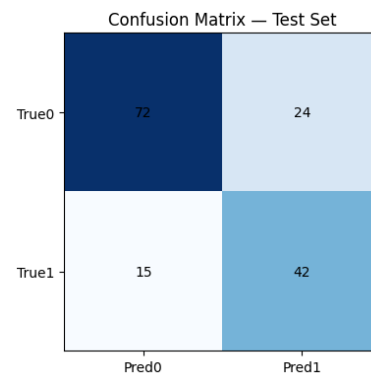
A high recall cannot show much when the model is not calibrated on confidence. AUC is used to direct the quality of decision separation and Experiment 3 showed better performance in ROC. Thus, regarding the aspect of medical safety, where missing diagnoses are undesirable, AUC would be required to inform the choice of models, and, consequently, Experiment 3 is regarded as the most suitable model to choose in order to screen diabetes.

3.2 Final Model Selection and Test Set Evaluation

Based on validation performance, Experiment 3 (Large model) was selected as the final model due to its superior clinical prioritization metrics. It was then tested on the unseen testing dataset consisting of 155 samples, producing the following results:

Table 2. Test Set Performance of Selected Final Model

Metric	Test Score	Interpretation
Accuracy	0.7451	Good overall classification
Precision	0.6364	Acceptable false-positive balance
Recall	0.7368	Strong diabetic detection
F1-score	0.6829	Balanced performance
AUC	0.8480	High reliability in probability ranking



3.3 Conclusion of Experimental Findings

The results of the evaluation reveal that AUC is the most valuable measure of clinical usefulness because it guarantees the consistency and reliability of separation between high-risk and low-risk groups. According to this measure, Experiment 3 was found to have the highest level of predictive confidence and generalization. Whereas the accuracy and the F1-score were a bit lower than in the smaller model, the performance was stable both in the validation and the test dataset. The big recall and AUC values on the test set (Recall = 0.7368, AUC = 0.8480) prove that the model can effectively classify the majority of diabetic patients. Thus, Experiment 3 was chosen as the final and most clinically appropriate architecture.

References

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